The instructions in the video for this particular lesson have been updated, please see the lesson notes below for the corrected version.

Learning Goals

- Loading audio files
- Playing audio files
- Using Audacity to export audio files

Let's get started by having a look at the folder structure.

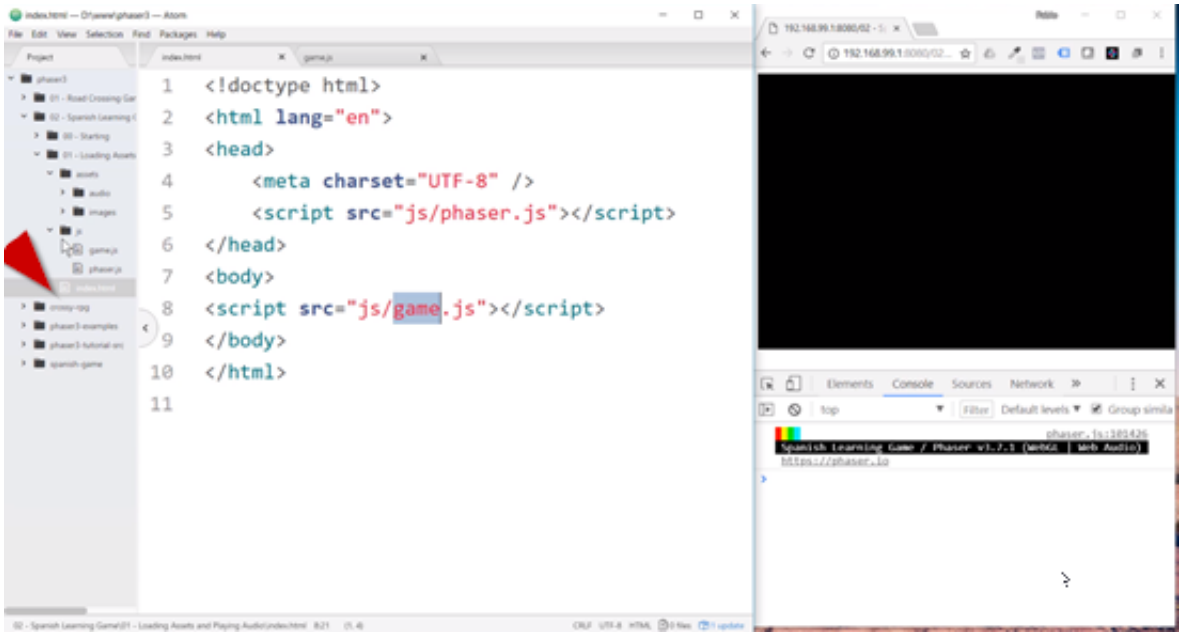
The file that is labeled **"00-Starting"** has all the assets and everything you need for this course.

This lesson will be coded in the folder **"01-Loading Assets."**

So if you want to follow along you should start from the 00-Starting folder.

The folder structure is as follows:

index.html which includes the Phaser library and the script for our game.



Then we have a folder named JS which contains the Phaser library and our JavaScript file. The JavaScript file will be used to code our game.

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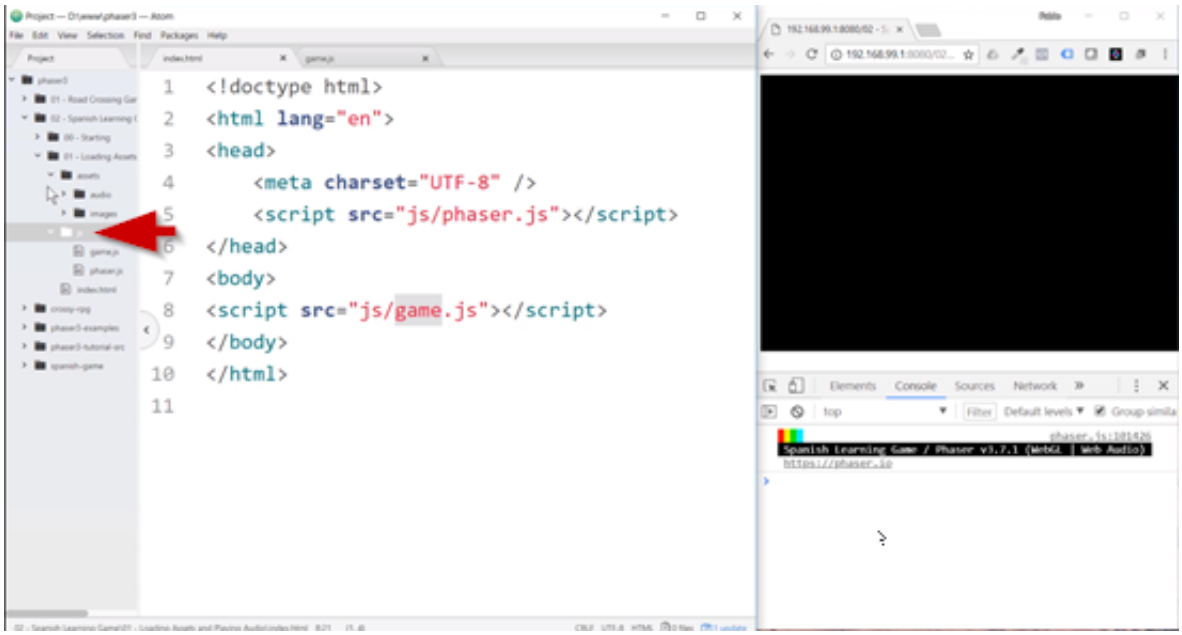
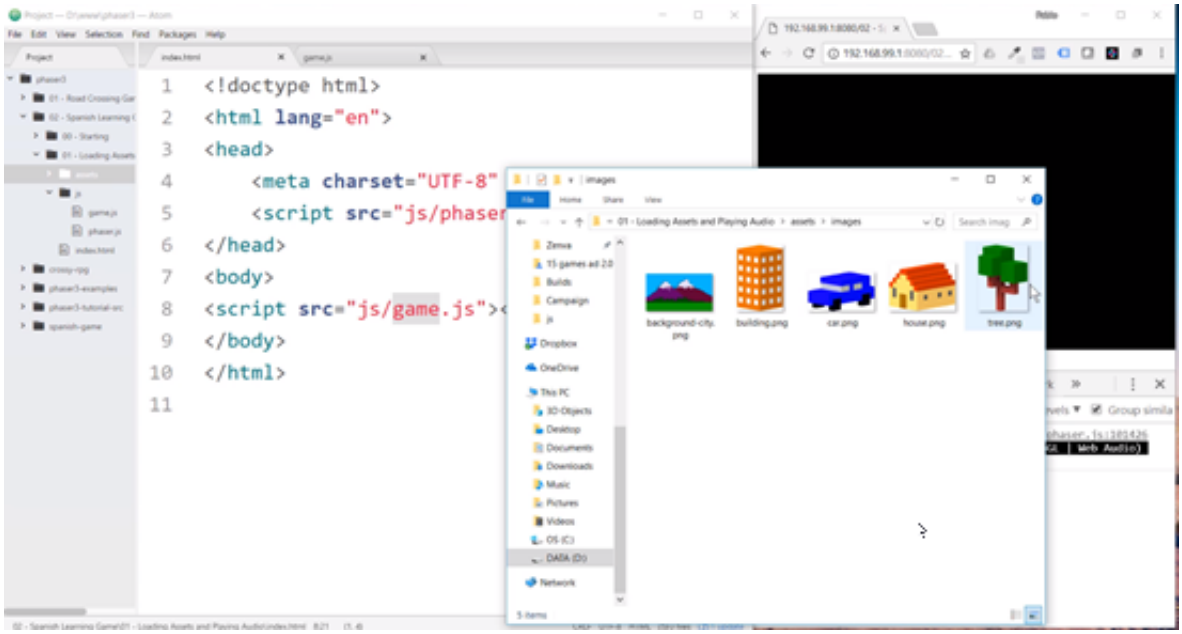


Then we have a **folder named Assets**, and in this folder we have a **sub folder called images** and in there we have the sprites that we will be using in this game.



Then we have the **sub folder called audio**, and in this folder we have the audio files we will use for our game.

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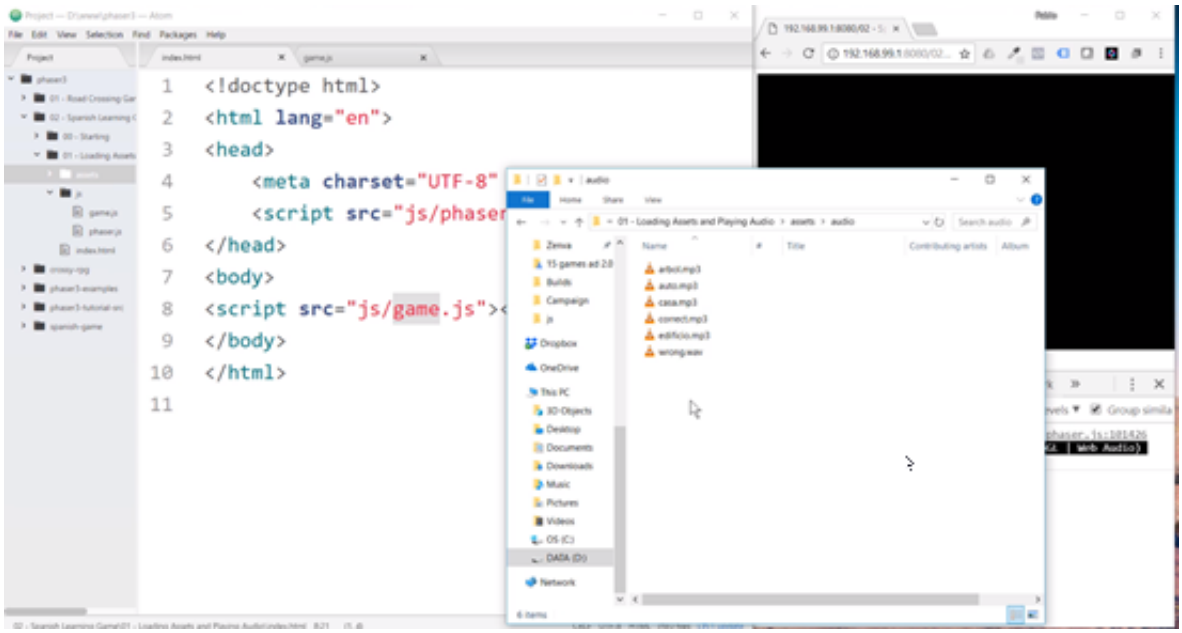
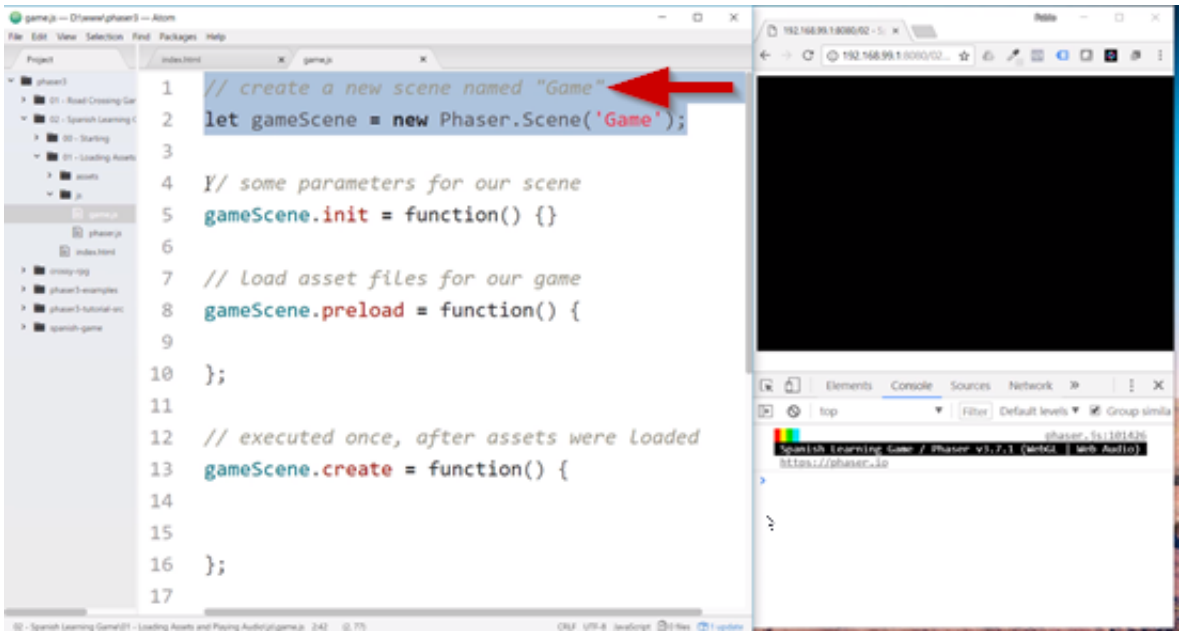


Open the game.js file and take a look, this file starts off by declaring the creation of a new scene named **"Game"**



Then we have the basic methods of our scene. So we have .init and we have .preload and we have .create.

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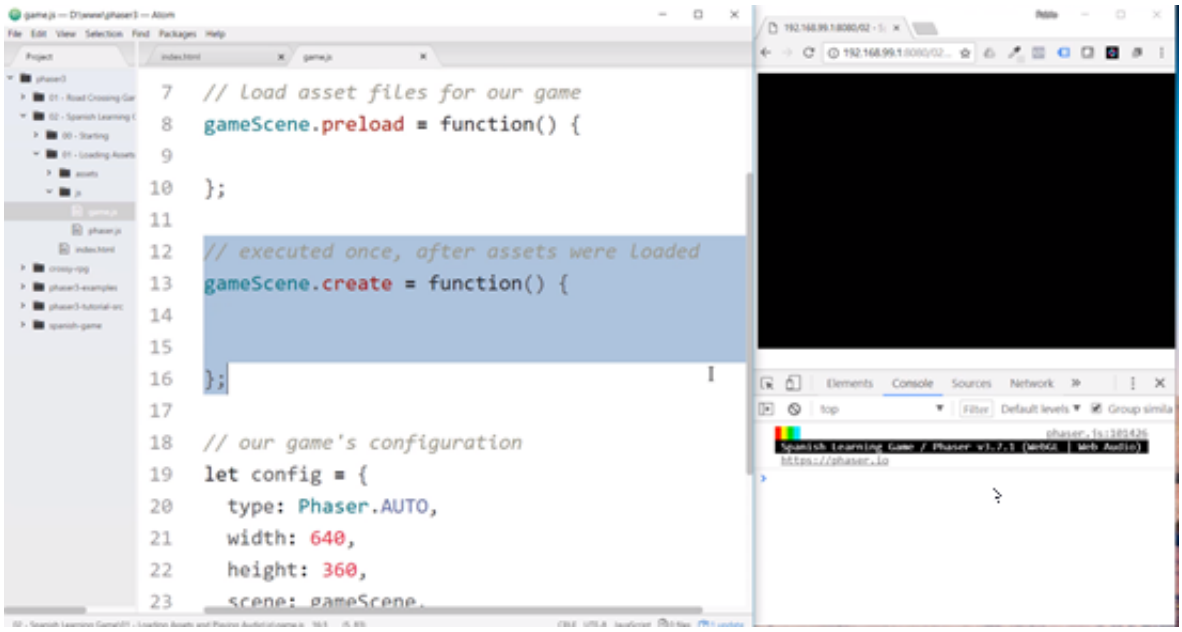
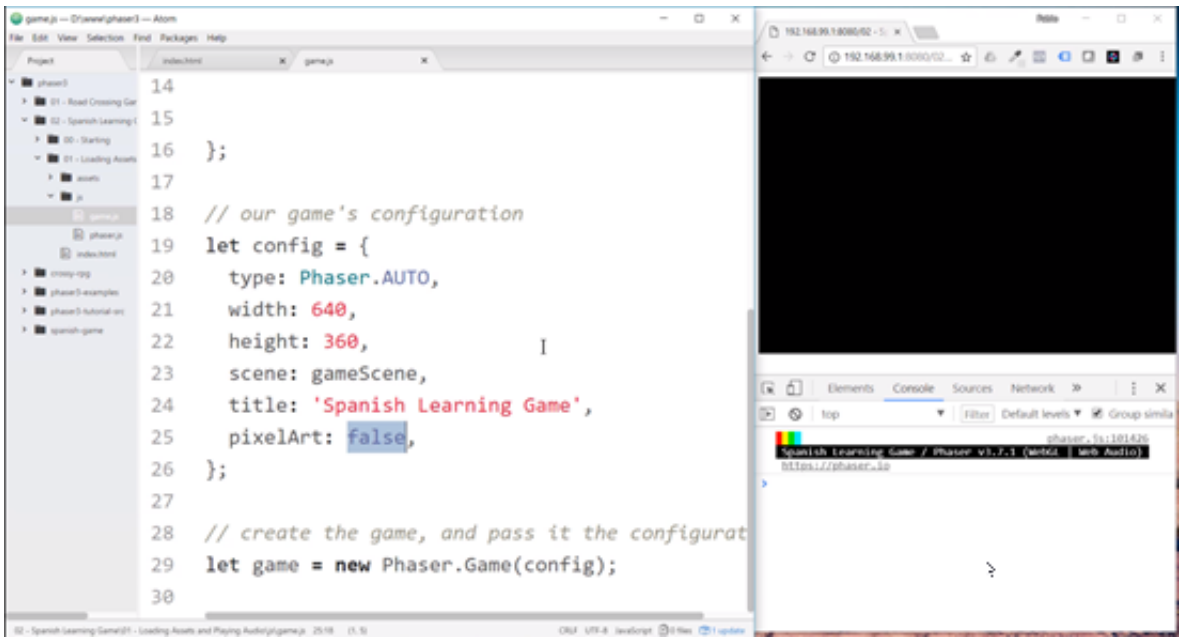


Then we have the configuration of our game which contains the size of the game, how we are rendering, which is with Auto as opposed to Canvas or WebGL. Then we pass on the scene that we created, setting the title of the game, and the pixelArt flag, this is used to turn whether you want the rendering to be pixel perfect.



Loading our Images & Audio

The first thing we are going to do is load our assets in a preload method. This will read all the assets from the disk and place them into the memory of our browser so that they are ready to go when our game begins. You would never want a game where the player is being loaded as you start playing. You want everything to be loaded first and when the game begins you want everything to be there on the screen.

See the code below and follow along:

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```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {
  // load background
  this.add.sprite(0, 0, 'background').setOrigin(0, 0);


};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

Make sure the **script is saved.**

Converting Audio Files

One of the audio files that we have is a .wav file and we need to change this to an MP3 file.

We can make this change to the file using a program that is free called Audacity.

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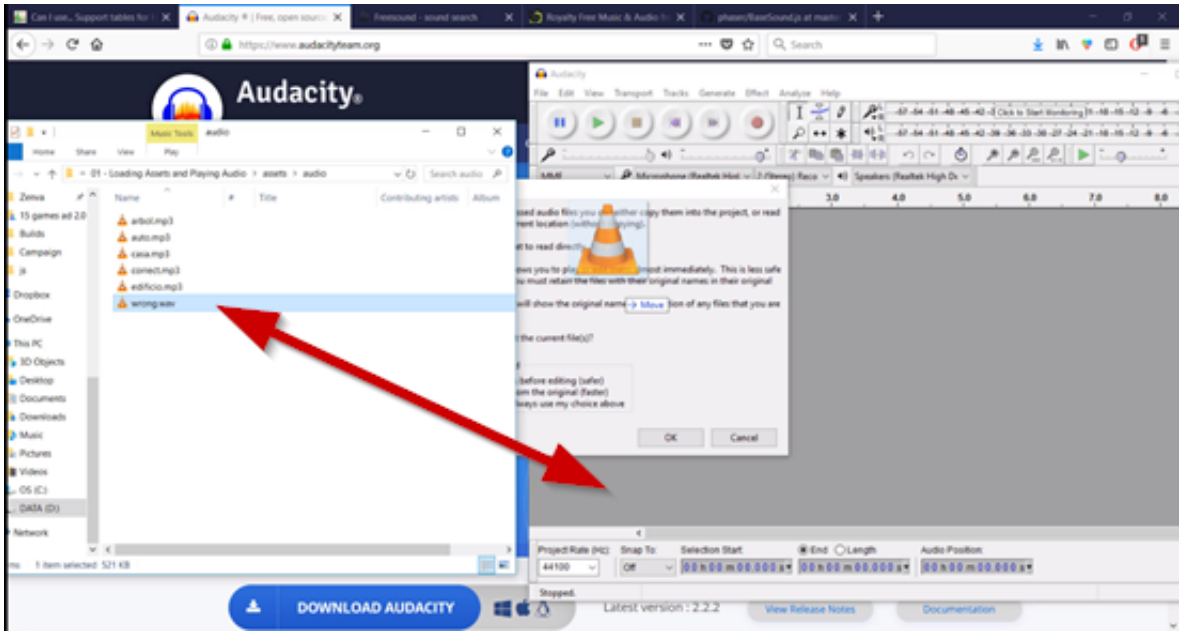
You can **download Audacity from here** : https://www.audacityteam.org/download/

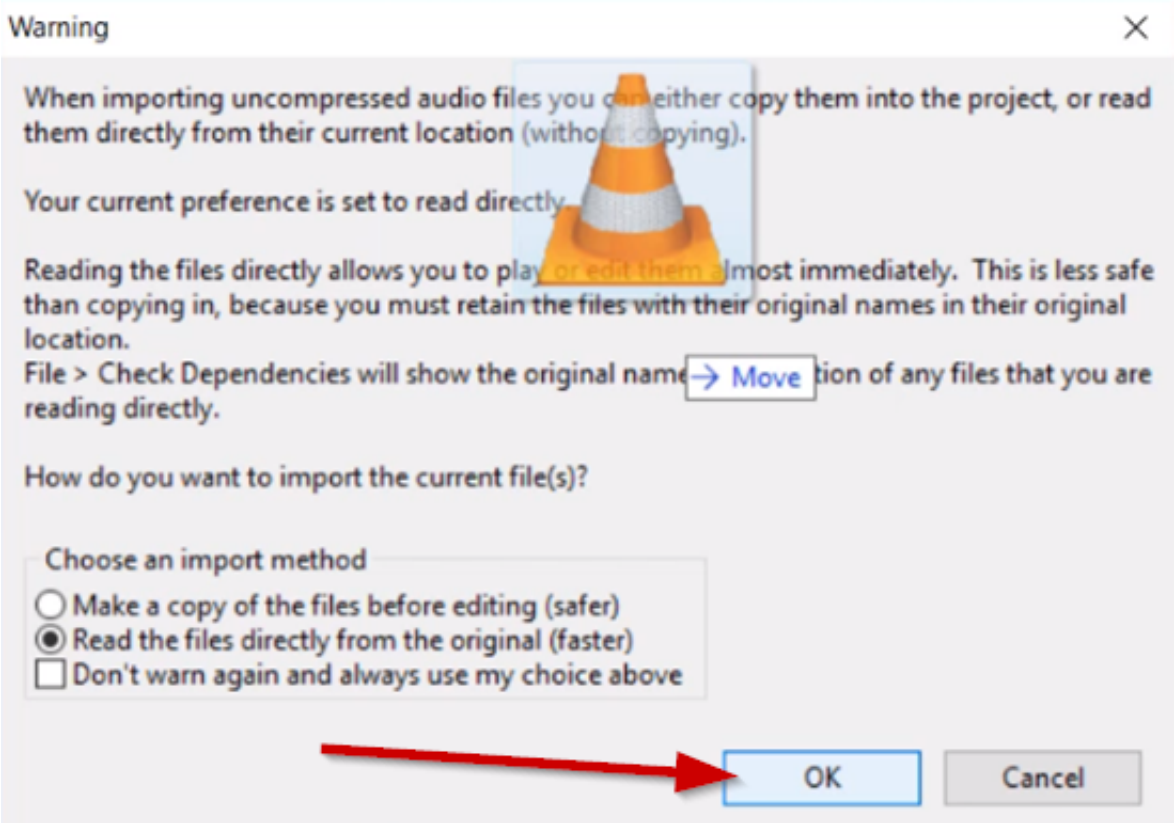
The direct link to download Audacity is here: Audacity

Once you have Audacity downloaded and installed go ahead and open the program.

You can select the file from your file explorer and drag it into the Audacity program.



You will be prompted with a Warning window. Select the second option which is the **Read the files directly from the original** then click the **OK button.**



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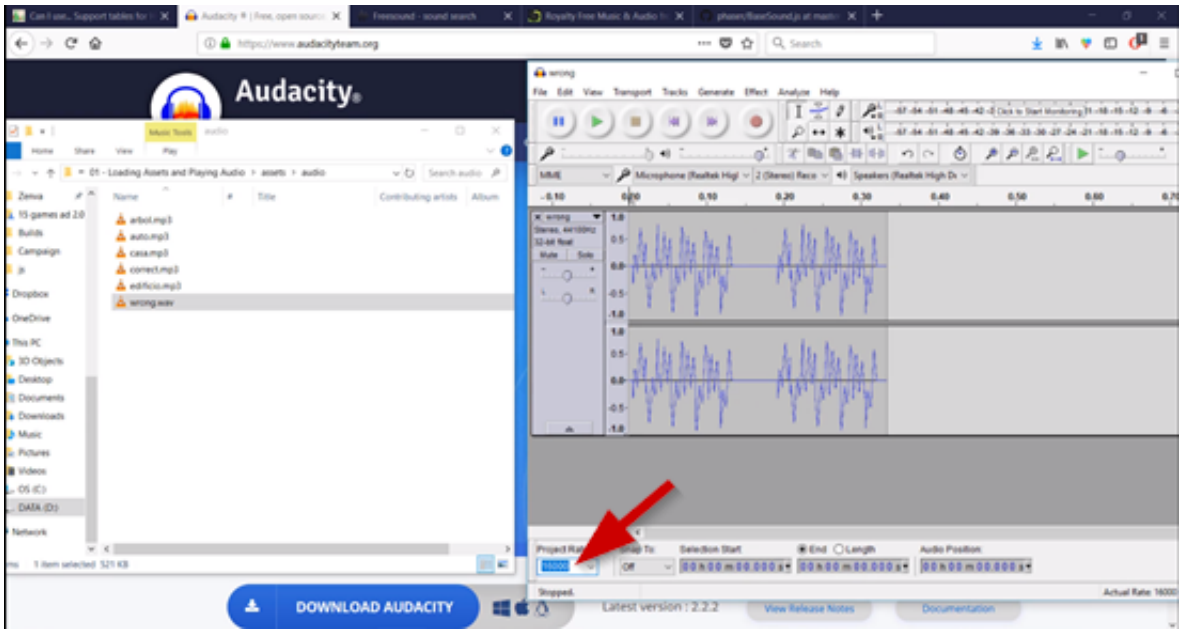
One of the things you will notice about the sound file is that it is quite long, and we can reduce the size of this file by **selecting the empty space and then hit the delete key to remove the empty space from the file.**

Now you can **reduce the Project Rate to 16000.**



Now you can export the sound file by going to **File>Export Audio, then find the location of the folder and select MP3 format then hit the Save button.**

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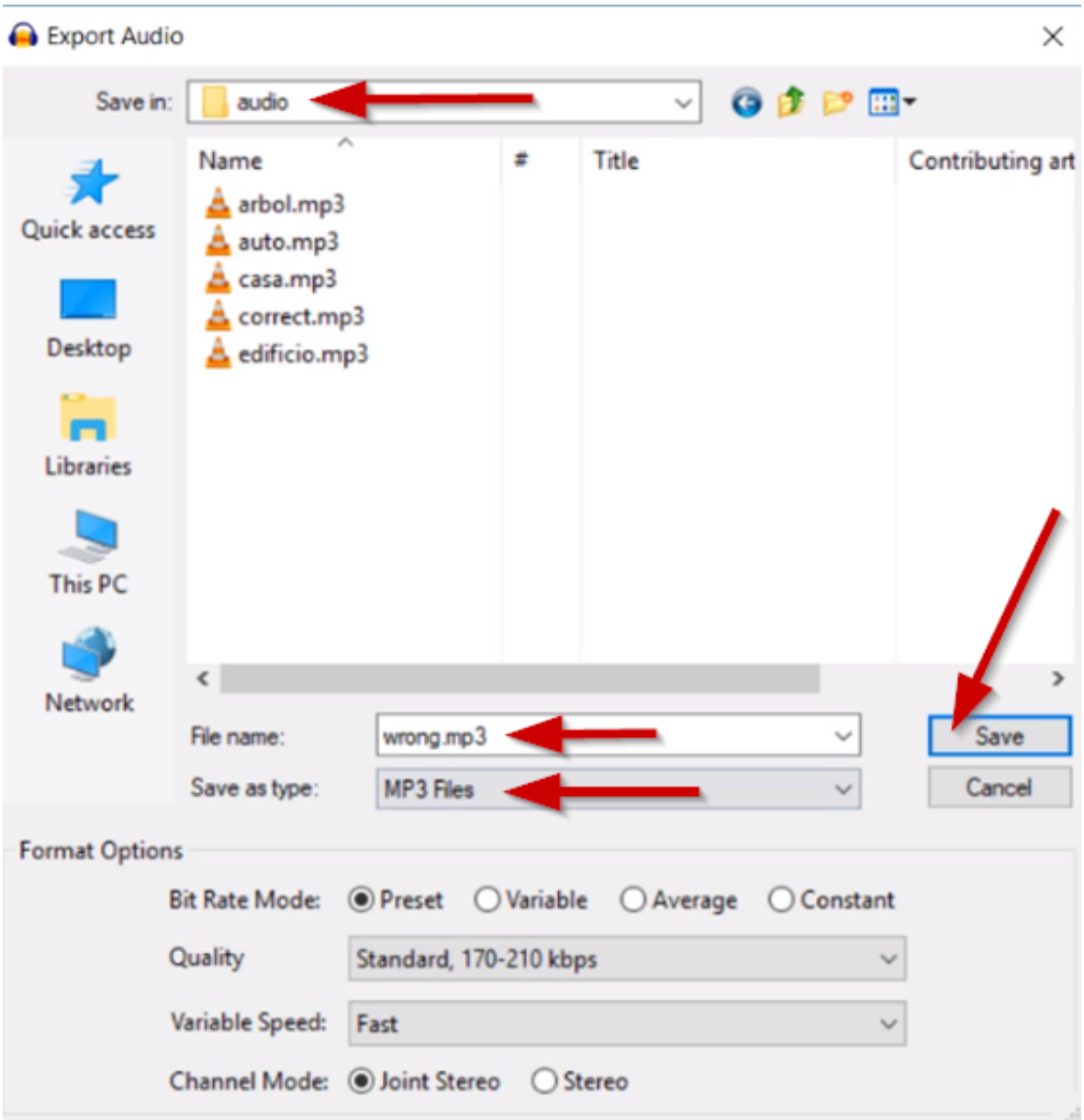

We now have the MP3 file and you can see that the size is now 5kb.

You can **delete the wrong.wav file** from the folder.

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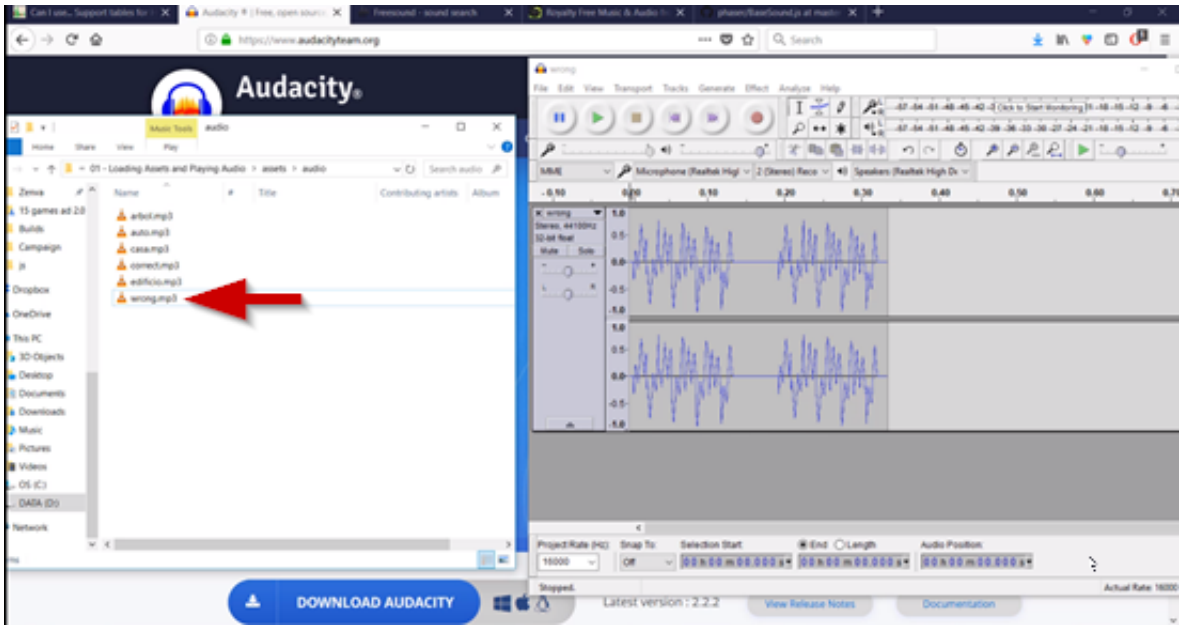

Playing Audio

You can play a file in Phaser so navigate back to the **game.js file** and we will play one of our audio files when the game starts.

For that got to .create and we need to create a sound object.

See the code below and follow along:

```
// executed once, after assets were loaded
gameScene.create = function() {
  // load background
  this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  let soundSample = this.sound.add('correct');
  soundSample.play();
  // soundSample.stop();
  // soundSample.pause();
  // soundSample.resume();

};
```

Save the file, and **refresh** the web page.

You will hear the sound play.

The following instructions have been updated, and differ from the video:

Please note that Google Chrome blocks the first sound playing until you interact with the screen (by clicking or pressing any keyboard key). So, if you don't hear any sound, ensure you click somewhere in the browser before investigating the audio playing code.

Lesson Summary

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Loading audio

- For the latest browser support of a format check **caniuse.com**
- MP3 has almost full support (except for Opera Mini)
- Loading audio is similar to loading sprites:

```
this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
```

Playing audio

- Create a **sound** object
- Useful methods: **play(), stop(), pause(), resume()**

```
let soundSample = this.sound.add('correct');
soundSample.play();
```

Playing audio

- Create a **sound** object
- Useful methods: **play(), stop(), pause(), resume()**

```
let soundSample = this.sound.add('correct');
soundSample.play();
```

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Code Correction in the Lesson Notes: The code in the video for this particular lesson has been updated, please see the lesson notes below for the corrected code.

Learning Goals

- Creating multiple sprites in a group
- Sorting group and sprite position using the **depth** property
- Making sprites interactive
- Listening to pointer down events

Before we make any modifications to the code we have already written, there is something that needs to be changed on a CSS level. We need to have the game width to fit 100% of the space it has available. This is so we can see everything when the screen is small.

Open the index.html file.

See the code below and follow along:

```
<!doctype html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <script src="js/phaser.js"></script>
    <style>
      canvas { width: 100%; }
    </style>
</head>
<body>
<script src="js/game.js"></script>
</body>
</html>
```

Save the index.html file and refresh the page.

What we need to do now is remove that default sound we have playing in the beginning, otherwise it will keep playing if we leave it.

Open the game.js file.

See the code below and follow along:

The following code has been updated, and differs from the video:

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
```

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```
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group([{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      }
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      }
    }
  ]);
```

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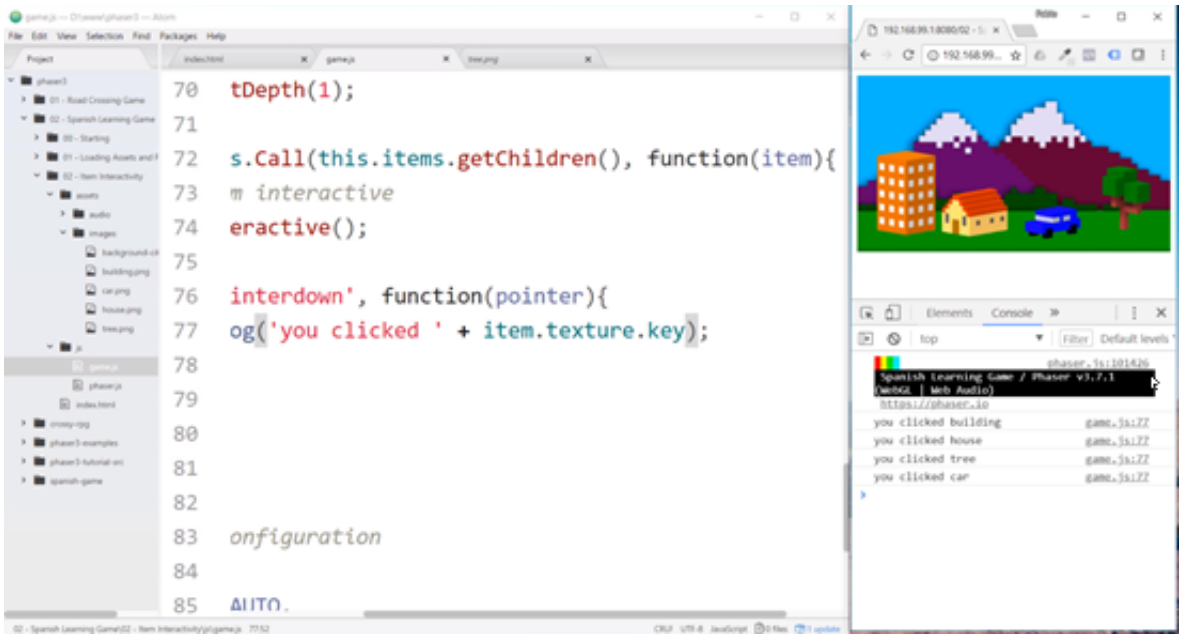
```
  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  // show group sprites on top of the background
  this.items.setDepth(1);

  Phaser.Actions.Call(this.items.getChildren(), function(item) {

    // make item interactive
    item.setInteractive();

    // listen to the pointerdown event
    item.on('pointerdown', function(pointer) {
      console.log('you clicked ' + item.texture.key);
    });
  }, this);

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

Save the game.js file and refresh the web page.



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Lesson Summary

Creating multiple sprites in a group

- this.group.add(/* array of objects */)

```
this.items = this.add.group([{
    key: 'building',
    setXY: {
      x: 100,
      y: 240
    },
    setScale: {
      x: 0.8,
      y: 0.8
    }
  }]);
```

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Depth property

- Sprites with higher depth are rendered on top of those with lower depth
- Setting depth for all group items:

```
this.items.setDepth(1);
```

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Interactive objects and events

- Enables the **Input** plugin on Phaser

```
// make item interactive
item.setInteractive();

// listen to the pointerdown event
item.on('pointerdown', function(pointer) {
  console.log('you clicked ' + item.texture.key);
});
```

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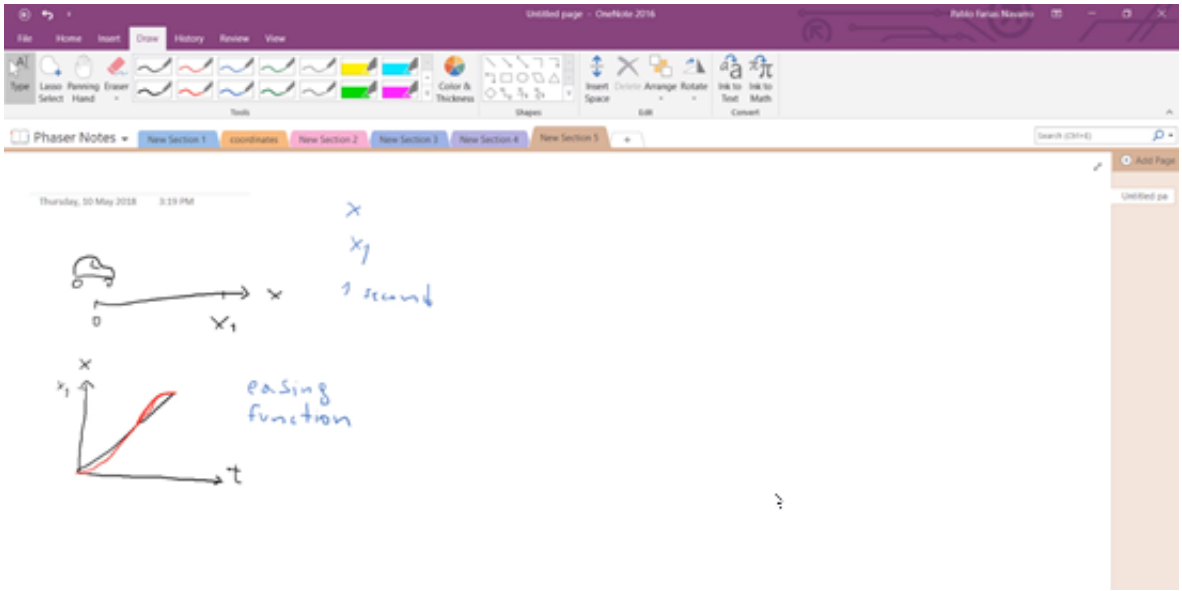
The code in the video for this particular lesson has been updated, please see the lesson notes below for the corrected code.

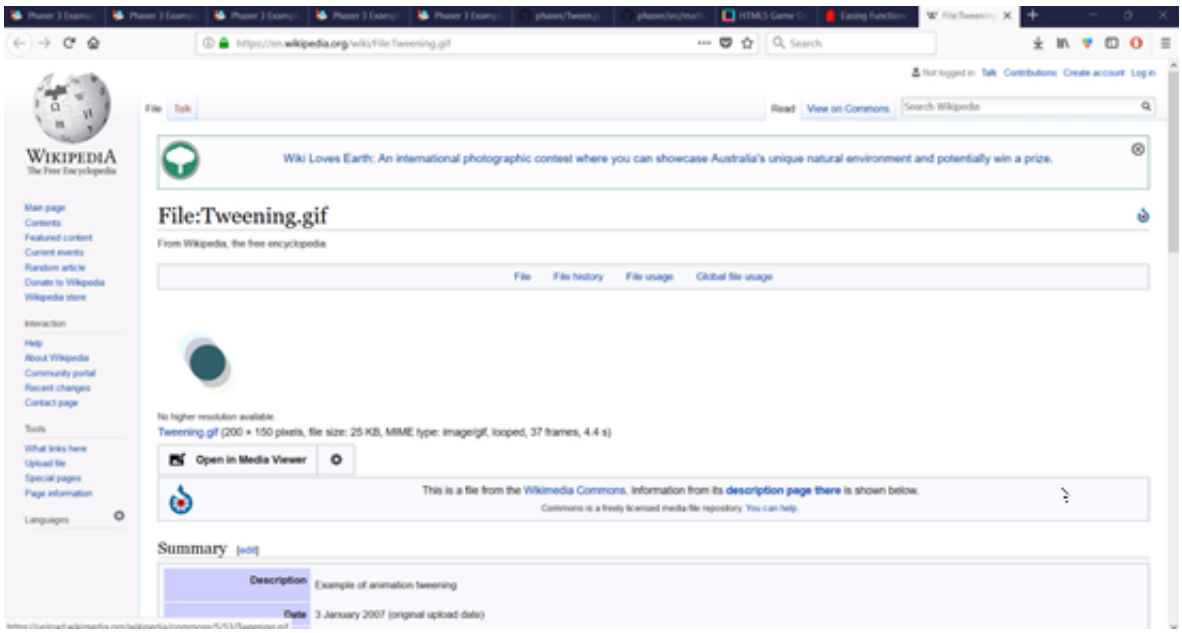
Learning Goals

- Easing functions and tweening concepts
- Tweening properties in Phaser
- Point over and out events

In this you will be introduced to the concept that is very important, not just in Phaser, but in game development and animation in general. This concept is called **“tweening”** so let’s begin by presenting the following simple example here:



There is an example from wikipedia that shows how this ball has different properties being tweened over time.



Here is the link to the wikipedia article: https://en.wikipedia.org/wiki/File:Tweening.gif

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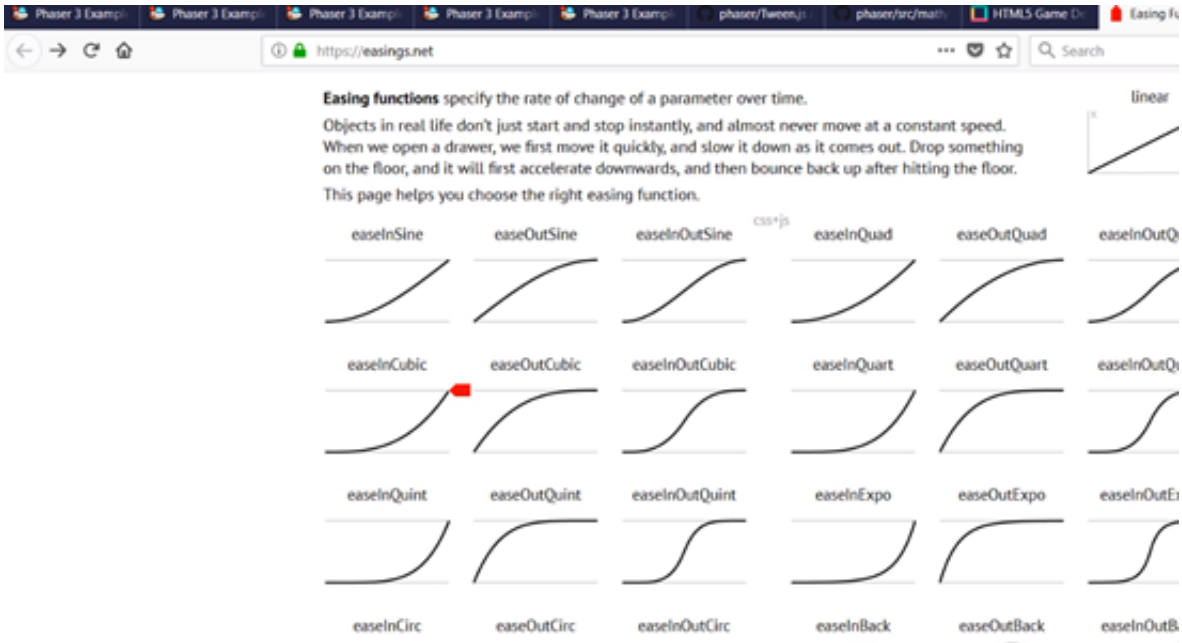
The direct link is here: Tweening

The website where you can obtain more information about curves is here: https://easing.net

Here is the direct link: Easing functions



We will now add some tween effects to our game that will respond to the input of the user.

Open the game.js file.

See the code below and follow along.

The following code has been updated, and differs from the video:. To work with more recent versions of **Phaser 3**, you need to use the method play() to start the tween animations instead of restart(), as used in the video. Stopping the tween with stop() is also no longer required. However, we'll also now need to ensure the tween is retained for future reuse by adding persist: true when adding the tween:

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
```

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```
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group([{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      }
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      }
    }
  ]);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  // show group sprites on top of the background
  this.items.setDepth(1);

  Phaser.Actions.Call(this.items.getChildren(), function(item) {
```

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```
    // make item interactive
    item.setInteractive();

    // creating tween - resize tween
    item.resizeTween = this.tweens.add({
      targets: item,
      scaleX: 1.5,
      scaleY: 1.5,
      duration: 300,
      paused: true,
      yoyo: true,
      ease: 'Quad.easeInOut',
      persist: true
    });

    // transparency tween
    item.alphaTween = this.tweens.add({
      targets: item,
      alpha: 0.7,
      duration: 200,
      paused: true,
      persist: true
    });

    // listen to the pointerdown event
    item.on('pointerdown', function(pointer) {
      item.resizeTween.play();
    });

    // listen to the pointerover event
    item.on('pointerover', function(pointer) {
      item.alphaTween.play();
    });

    // listen to the pointerout event
    item.on('pointerout', function(pointer) {
      // set no transparency
      item.alpha = 1;
    });

  }, this);

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
```

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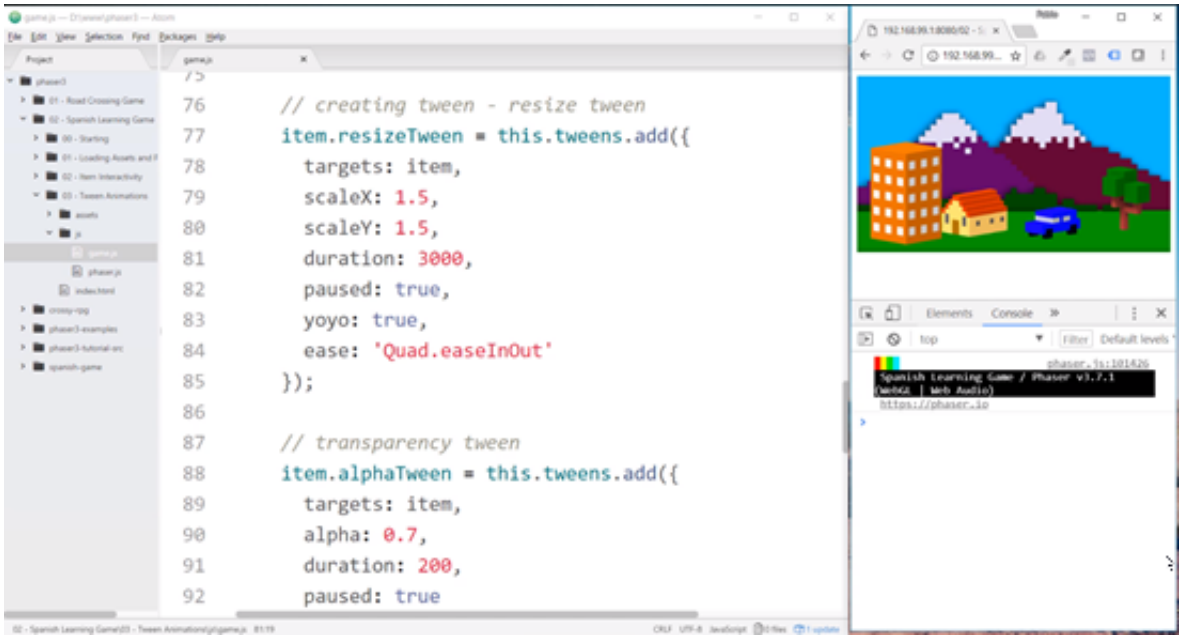
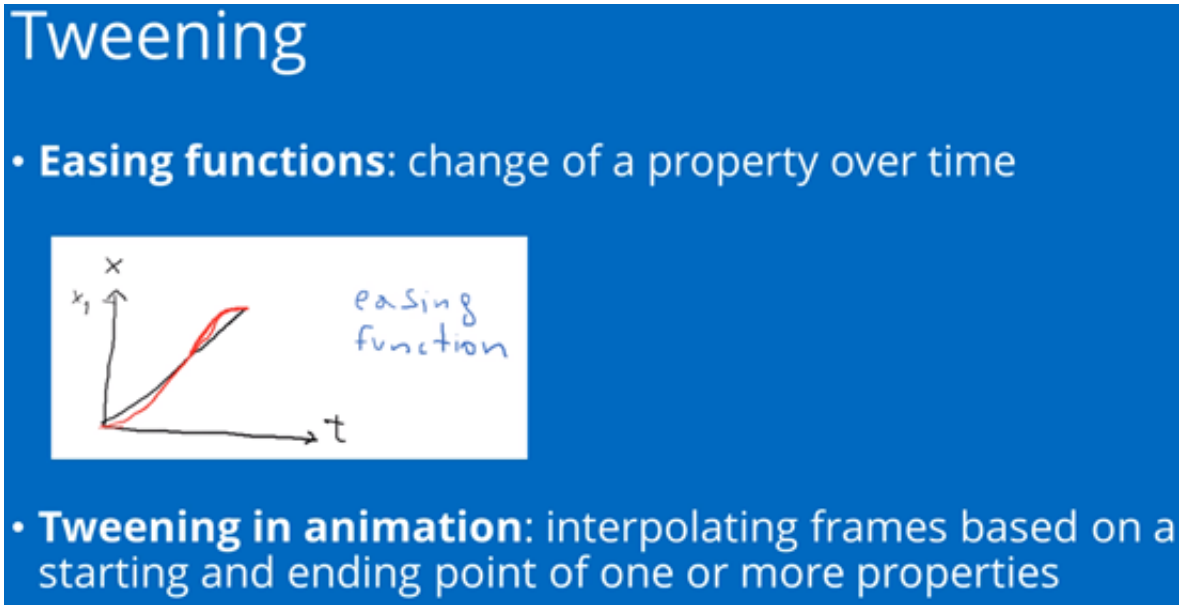
```
let game = new Phaser.Game(config);
```

Save the game.js file and refresh the web page.



Lesson Summary



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Tweening in Phaser

```
// creating tween - resize tween
item.resizeTween = this.tweens.add({
  targets: item,
  scaleX: 1.5,
  scaleY: 1.5,
  duration: 300,
  paused: true,
  yoyo: true,
  ease: 'Quad.easeInOut'
});
```

```
item.resizeTween.restart();
```

```
item.alphaTween.stop();
```

Point over and out events

- Triggered on interactive objects

```
// listen to the pointerover event
item.on('pointerover', function(pointer) {
  item.alphaTween.restart();
});

// listen to the pointerout event
item.on('pointerout', function(pointer) {
  //stop alpha tween
  item.alphaTween.stop();

  // set no transparency
  item.alpha = 1;
});
```

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Learning Goals

- Showing text
- Basic question logic
- Picking a random element from an array

What we need to do now is concentrate a bit on the game logic. We need to have a method that we will call **“Show next question”** when this method is called which will be at the beginning and after the user gives a response, we want a new Spanish word to appear up at the top of the game window, and we need the sound of that word , in Spanish, to be played. So the player will hear a word and see it written and then the Player will have to select to which element it corresponds to.

Open the game.js file.

See the code below and follow along:

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {
  // word database
  this.words = [{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      },
      spanish: 'edificio'
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'casa'
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'automóvil'
    },
    {
```

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```
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      },
      spanish: 'árbol'
    }
  ];
}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group(this.words);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  // show group sprites on top of the background
  this.items.setDepth(1);

  // getting group array
  let items = this.items.getChildren();

  for(let i = 0; i < items.length; i++) {

    let item = items[i];

    // make item interactive
    item.setInteractive();

    // creating tween - resize tween
    item.resizeTween = this.tweens.add({
      targets: item,
      scaleX: 1.5,
      scaleY: 1.5,
      duration: 300,
      paused: true,
```

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```
      yoyo: true,
      ease: 'Quad.easeInOut',
      persist: true
    });

    // transparency tween
    item.alphaTween = this.tweens.add({
      targets: item,
      alpha: 0.7,
      duration: 200,
      paused: true,
      persist: true
    });

    // listen to the pointerdown event
    item.on('pointerdown', function(pointer) {
      item.resizeTween.play();

      // show next question
      this.showNextQuestion();
    }, this);

    // listen to the pointerover event
    item.on('pointerover', function(pointer) {
      item.alphaTween.play();
    }, this);

    // listen to the pointerout event
    item.on('pointerout', function(pointer) {
      // set no transparency
      item.alpha = 1;
    }, this);

    // create sound for each word
    this.words[i].sound = this.sound.add(this.words[i].key + 'Audio');

  }

  // text object
  this.wordText = this.add.text(30, 20, ' ', {
    font: '28px Open Sans',
    fill: '#ffffff'
  });

  // show the first question
  this.showNextQuestion();
};

// show new question
gameScene.showNextQuestion = function(){

  // select a random word
  let nextWord = Phaser.Math.RND.pick(this.words);

  // play a sound for that word
```

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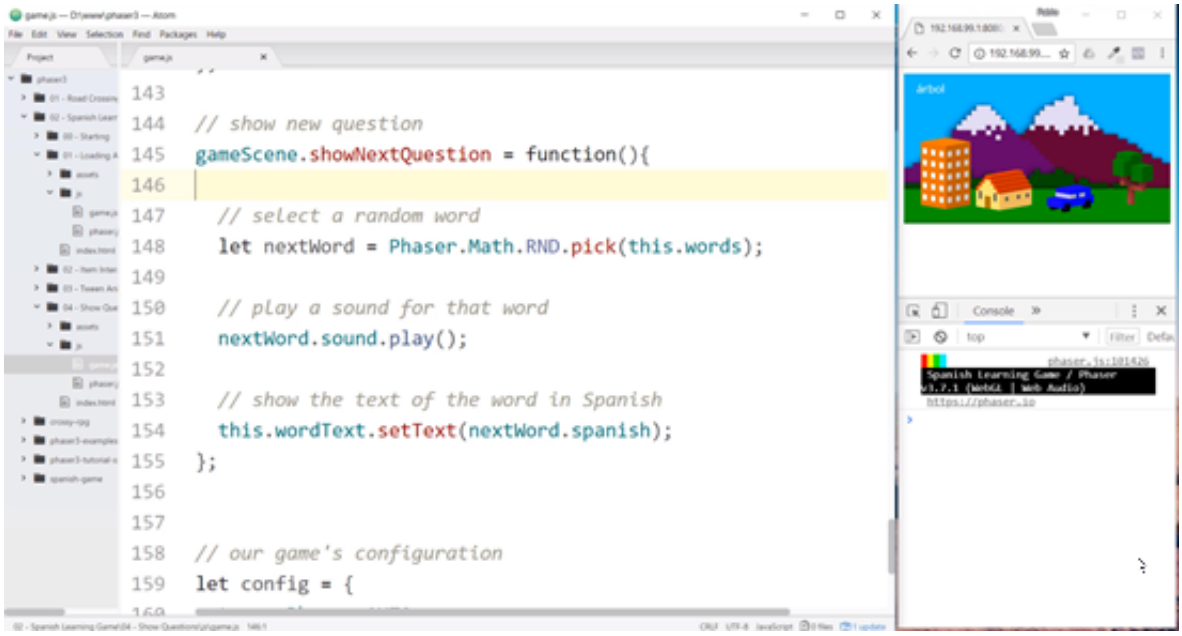
```
  nextWord.sound.play();

  // show the text of the word in Spanish
  this.wordText.setText(nextWord.spanish);
};


// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

Save the game.js file and refresh the web page.



Lesson Summary

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Showing text

- Create a text object:

```
// text object
this.wordText = this.add.text(30, 20, 'hi there', {
  font: '28px Open Sans',
  fill: '#ffffff'
});
```

- Change text: **this.wordText.setText("Hello")**
- Origin is in the center

Question logic

1. Pick a random question

```
// select a random word
let nextWord = Phaser.Math.RND.pick(this.words);
```

2. Play sound

```
// play a sound for that word
nextWord.sound.play();
```

3. Show text

```
// show the text of the word in Spanish
this.wordText.setText(nextWord.spanish);
```

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The instructions in the video for this particular lesson have been updated, please see the lesson notes below for the corrected code.

Learning Goals

- Checking responses logic
- Play a sound depending on the result
- Different tween effects for right/wrong answers

We want to be able to check the answer of the user and to let them know whether they go it correct or incorrect. So, we will need to **create a method** for this. This method will be called **"checkAnswer."**

Open the game.js file.

See the code below and follow along:

```
      },
      spanish: 'automóvil'
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      },
      spanish: 'árbol'
    }
  ];
}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group(this.words);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);
```

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```
// show group sprites on top of the background
this.items.setDepth(1);

// getting group array
let items = this.items.getChildren();

for(let i = 0; i < items.length; i++) {

  let item = items[i];

  // make item interactive
  item.setInteractive();

  // creating tween - resize tween
  item.correctTween = this.tweens.add({
    targets: item,
    scaleX: 1.5,
    scaleY: 1.5,
    duration: 300,
    paused: true,
    yoyo: true,
    ease: 'Quad.easeInOut',
    persist: true
  });

  item.wrongTween = this.tweens.add({
    targets: item,
    scaleX: 1.5,
    scaleY: 1.5,
    duration: 300,
    angle: 90,
    paused: true,
    yoyo: true,
    ease: 'Quad.easeInOut',
    persist: true
  });

  // transparency tween
  item.alphaTween = this.tweens.add({
    targets: item,
    alpha: 0.7,
    duration: 200,
    paused: true,
    persist: true
  });

  // listen to the pointerdown event
  item.on('pointerdown', function(pointer) {

    let result = this.processAnswer(this.words[i].spanish);

    // depending on the result, we'll play one tween or the other
    if(result) {
      item.correctTween.play();
```

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```
      }
      else {
        item.wrongTween.play();
      }

      // show next question
      this.showNextQuestion();
    }, this);

    // listen to the pointerover event
    item.on('pointerover', function(pointer) {
      item.alphaTween.play();
    }, this);

    // listen to the pointerout event
    item.on('pointerout', function(pointer) {
      // set no transparency
      item.alpha = 1;
    }, this);

    // create sound for each word
    this.words[i].sound = this.sound.add(this.words[i].key + 'Audio');

  }

  // text object
  this.wordText = this.add.text(30, 20, ' ', {
    font: '28px Open Sans',
    fill: '#ffffff'
  });

  // correct / wrong sounds
  this.correctSound = this.sound.add('correct');
  this.wrongSound = this.sound.add('wrong');

  // show the first question
  this.showNextQuestion();
};

// show new question
gameScene.showNextQuestion = function(){

  // select a random word
  this.nextWord = Phaser.Math.RND.pick(this.words);

  // play a sound for that word
  this.nextWord.sound.play();

  // show the text of the word in Spanish
  this.wordText.setText(this.nextWord.spanish);
};

// answer processing
gameScene.processAnswer = function(userResponse) {
```

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```
  // compare user response with correct response
  if(userResponse == this.nextWord.spanish) {
    // it's correct

    // play sound
    this.correctSound.play();

    return true;
  }
  else {
    // it's wrong

    // play sound
    this.wrongSound.play();

    return false;
  }
}

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

The following instructions have been updated, and differs from the video:

Save the game.js file and refresh the web page. Be aware that Google Chrome blocks the first sound until you interact with the screen. So you need to click on the screen or press any key before it plays any sounds in the game.

The game is running correctly.

Congratulations you have completed this course and have created a Spanish Learning Game.

Lesson Summary

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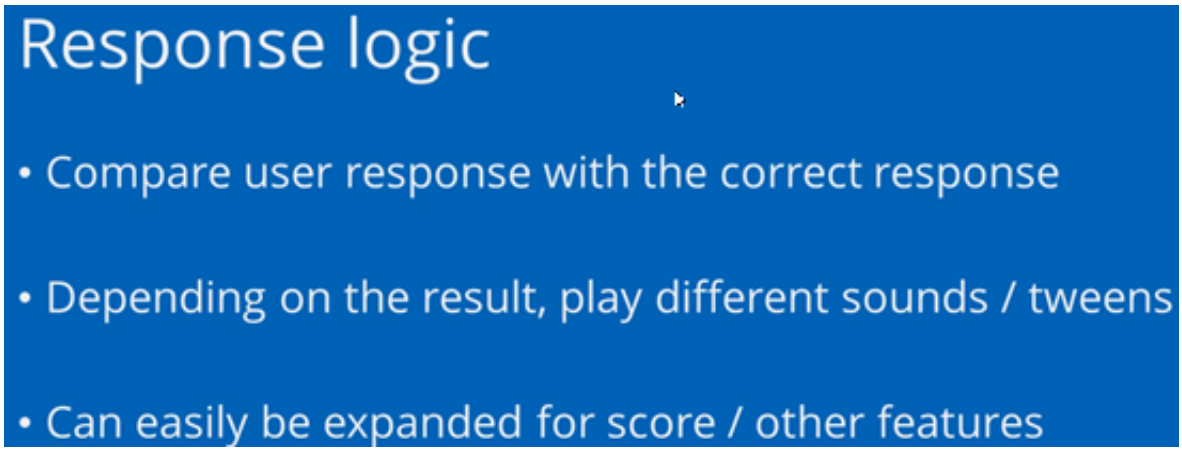

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In this lesson, you can find the full source code for the project used within the course. Feel free to use this lesson as needed – whether to help you debug your code, use it as a reference later, or expand the project to practice your skills!

You can also download all project files from the **Course Files** tab via the project files or downloadable PDF summary.

game.js

Found in Project/00 – Starting/js

This code sets up a Phaser-based game with a single scene “Game”. The game parameters and assets to be used in the scene are defined, with functions for initialization and creation of the game elements. A Phaser game instance is then created with the specified game configuration details.

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {

};

// executed once, after assets were loaded
gameScene.create = function() {

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

game.js

Found in Project/01 – Loading Assets and Playing Audio/js

This code creates a Phaser-based Spanish Learning Game with one scene named “Game”. It initiates the game parameters, preloads required images and audio assets, generates a background on creation, and plays a sound sample. Finally, it sets the game configurations before creating the Phaser game instance.

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```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {
  // load background
  this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  let soundSample = this.sound.add('correct');
  soundSample.play();
  // soundSample.stop();
  // soundSample.pause();
  // soundSample.resume();

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

game.js

Found in Project/02 – Item Interactivity/js

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This code creates a Phaser-based Spanish Learning Game with image and audio assets. Once assets are loaded, interactive items are created in the scene and placed at specific coordinates on top of a background image. When individual items are clicked, a console log message is displayed with the item's name. The game configuration is then set followed by creating a Phaser game instance with that configuration.

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group([{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      }
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
```

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```
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      }
    }
  ]);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  // show group sprites on top of the background
  this.items.setDepth(1);

  Phaser.Actions.Call(this.items.getChildren(), function(item) {

    // make item interactive
    item.setInteractive();

    // listen to the pointerdown event
    item.on('pointerdown', function(pointer) {
      console.log('you clicked ' + item.texture.key);
    });
  }, this);

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

game.js

Found in Project/03 – Tween Animations/js

The provided code creates a Phaser-based game, where images are interactively displayed over a background. Once assets are loaded, items, with resizing and transparency tweens, are created and

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made interactive. On mouse events including click, mouse-over, and mouse-out, the items respond by playing, pausing, stopping, or reverting respective tweens. After setting the game configurations, a Phaser game instance is created.

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group([{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      }
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      }
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
```

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```
      x: 0.8,
      y: 0.8
    }
  },
  {
    key: 'tree',
    setXY: {
      x: 550,
      y: 250
    }
  }
]);

// background
let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

// show group sprites on top of the background
this.items.setDepth(1);

Phaser.Actions.Call(this.items.getChildren(), function(item) {

  // make item interactive
  item.setInteractive();

  // creating tween - resize tween
  item.resizeTween = this.tweens.add({
    targets: item,
    scaleX: 1.5,
    scaleY: 1.5,
    duration: 300,
    paused: true,
    yoyo: true,
    ease: 'Quad.easeInOut'
  });

  // transparency tween
  item.alphaTween = this.tweens.add({
    targets: item,
    alpha: 0.7,
    duration: 200,
    paused: true
  });

  // listen to the pointerdown event
  item.on('pointerdown', function(pointer) {
    item.resizeTween.play();
  });

  // listen to the pointerover event
  item.on('pointerover', function(pointer) {
    item.alphaTween.play();
  });

  // listen to the pointerout event
  item.on('pointerout', function(pointer) {
```

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```
      //stop alpha tween
      item.alphaTween.stop();

      // set no transparency
      item.alpha = 1;
    });

  }, this);

};

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

game.js

Found in Project/04 – Show Questions/js

This Phaser-based code creates a Spanish Learning Game where interactive images are displayed onto a scene. Each image is associated with a corresponding Spanish word and its sound. Upon clicking an image, the image expands, the associated word’s sound is played, and a new question is loaded. The Spanish word for the current item is also displayed on the screen. A Phaser game instance is then created with the predefined configuration.

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');

// some parameters for our scene
gameScene.init = function() {
  // word database
  this.words = [{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      },
      spanish: 'edificio'
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
```

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```
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'casa'
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'automóvil'
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      },
      spanish: 'árbol'
    }
  ];
}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');

  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group(this.words);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);
```

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```
// show group sprites on top of the background
this.items.setDepth(1);

// getting group array
let items = this.items.getChildren();

for(let i = 0; i < items.length; i++) {

  let item = items[i];

  // make item interactive
  item.setInteractive();

  // creating tween - resize tween
  item.resizeTween = this.tweens.add({
    targets: item,
    scaleX: 1.5,
    scaleY: 1.5,
    duration: 300,
    paused: true,
    yoyo: true,
    ease: 'Quad.easeInOut'
  });

  // transparency tween
  item.alphaTween = this.tweens.add({
    targets: item,
    alpha: 0.7,
    duration: 200,
    paused: true
  });

  // listen to the pointerdown event
  item.on('pointerdown', function(pointer) {
    item.resizeTween.play();

    // show next question
    this.showNextQuestion();
  }, this);

  // listen to the pointerover event
  item.on('pointerover', function(pointer) {
    item.alphaTween.play();
  }, this);

  // listen to the pointerout event
  item.on('pointerout', function(pointer) {
    //stop alpha tween
    item.alphaTween.stop();

    // set no transparency
    item.alpha = 1;
  }, this);

  // create sound for each word
```

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```
    this.words[i].sound = this.sound.add(this.words[i].key + 'Audio');

  }

  // text object
  this.wordText = this.add.text(30, 20, ' ', {
    font: '28px Open Sans',
    fill: '#ffffff'
  });

  // show the first question
  this.showNextQuestion();
};

// show new question
gameScene.showNextQuestion = function(){

  // select a random word
  let nextWord = Phaser.Math.RND.pick(this.words);

  // play a sound for that word
  nextWord.sound.play();

  // show the text of the word in Spanish
  this.wordText.setText(nextWord.spanish);
};


// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

game.js

Found in Project/05 – Checking Questions/js

This is a Spanish Learning Game built on Phaser. It displays images and plays the Spanish pronunciation for each image. When a user clicks on an image, the response is processed and based on correctness, a corresponding feedback sound is played and tweens are executed. The game moves on to the next question after each user interaction. It creates a Phaser game instance with prescribed configurations after initializing the entire setup.

```
// create a new scene named "Game"
let gameScene = new Phaser.Scene('Game');
```

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```
// some parameters for our scene
gameScene.init = function() {
  // word database
  this.words = [{
      key: 'building',
      setXY: {
        x: 100,
        y: 240
      },
      spanish: 'edificio'
    },
    {
      key: 'house',
      setXY: {
        x: 240,
        y: 280
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'casa'
    },
    {
      key: 'car',
      setXY: {
        x: 400,
        y: 300
      },
      setScale: {
        x: 0.8,
        y: 0.8
      },
      spanish: 'automóvil'
    },
    {
      key: 'tree',
      setXY: {
        x: 550,
        y: 250
      },
      spanish: 'árbol'
    }
  ];
}

// load asset files for our game
gameScene.preload = function() {
  // load images
  this.load.image('background', 'assets/images/background-city.png');
  this.load.image('building', 'assets/images/building.png');
  this.load.image('car', 'assets/images/car.png');
  this.load.image('house', 'assets/images/house.png');
  this.load.image('tree', 'assets/images/tree.png');
```

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```
  this.load.audio('treeAudio', 'assets/audio/arbol.mp3');
  this.load.audio('carAudio', 'assets/audio/auto.mp3');
  this.load.audio('houseAudio', 'assets/audio/casa.mp3');
  this.load.audio('buildingAudio', 'assets/audio/edificio.mp3');
  this.load.audio('correct', 'assets/audio/correct.mp3');
  this.load.audio('wrong', 'assets/audio/wrong.mp3');
};

// executed once, after assets were loaded
gameScene.create = function() {

  this.items = this.add.group(this.words);

  // background
  let bg = this.add.sprite(0, 0, 'background').setOrigin(0, 0);

  // show group sprites on top of the background
  this.items.setDepth(1);

  // getting group array
  let items = this.items.getChildren();

  for(let i = 0; i < items.length; i++) {

    let item = items[i];

    // make item interactive
    item.setInteractive();

    // creating tween - resize tween
    item.correctTween = this.tweens.add({
      targets: item,
      scaleX: 1.5,
      scaleY: 1.5,
      duration: 300,
      paused: true,
      yoyo: true,
      ease: 'Quad.easeInOut'
    });

    item.wrongTween = this.tweens.add({
      targets: item,
      scaleX: 1.5,
      scaleY: 1.5,
      duration: 300,
      angle: 90,
      paused: true,
      yoyo: true,
      ease: 'Quad.easeInOut'
    });

    // transparency tween
    item.alphaTween = this.tweens.add({
      targets: item,
```

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```
      alpha: 0.7,
      duration: 200,
      paused: true
    });

    // listen to the pointerdown event
    item.on('pointerdown', function(pointer) {

      let result = this.processAnswer(this.words[i].spanish);

      // depending on the result, we'll play one tween or the other
      if(result) {
        item.correctTween.play();
      }
      else {
        item.wrongTween.play();
      }

      // show next question
      this.showNextQuestion();
    }, this);

    // listen to the pointerover event
    item.on('pointerover', function(pointer) {
      item.alphaTween.play();
    }, this);

    // listen to the pointerout event
    item.on('pointerout', function(pointer) {
      //stop alpha tween
      item.alphaTween.stop();

      // set no transparency
      item.alpha = 1;
    }, this);

    // create sound for each word
    this.words[i].sound = this.sound.add(this.words[i].key + 'Audio');

  }

  // text object
  this.wordText = this.add.text(30, 20, ' ', {
    font: '28px Open Sans',
    fill: '#ffffff'
  });

  // correct / wrong sounds
  this.correctSound = this.sound.add('correct');
  this.wrongSound = this.sound.add('wrong');

  // show the first question
  this.showNextQuestion();
};
```

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```
// show new question
gameScene.showNextQuestion = function(){

  // select a random word
  this.nextWord = Phaser.Math.RND.pick(this.words);

  // play a sound for that word
  this.nextWord.sound.play();

  // show the text of the word in Spanish
  this.wordText.setText(this.nextWord.spanish);
};

// answer processing
gameScene.processAnswer = function(userResponse) {

  // compare user response with correct response
  if(userResponse == this.nextWord.spanish) {
    // it's correct

    // play sound
    this.correctSound.play();

    return true;
  }
  else {
    // it's wrong

    // play sound
    this.wrongSound.play();

    return false;
  }
}

// our game's configuration
let config = {
  type: Phaser.AUTO,
  width: 640,
  height: 360,
  scene: gameScene,
  title: 'Spanish Learning Game',
  pixelArt: false,
};

// create the game, and pass it the configuration
let game = new Phaser.Game(config);
```

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